Statistical mechanics

1st HW

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1. Explain the ideal gas system and the absolute temperature defined in thermodynamics. Find the answer in your study with reference books.

5/10

Ideal gas system is composed of extremly large number of ideal particles confined to a space.

"Ideal particle" have three properties.

① There are no potential between particles.

② Particles are small enough to ignore.

③ They always do elastic collisions.

Absolute temperature is a temperature scale fixed to absolute zero, measuring internal thermal kinetic energy

Absolute temperature is directly proportional to the average translational kinetic energy of ideal particles.

(in 3D,) $\left\langle \frac{1}{2} m v^2 \right\rangle = \frac{3}{2} k_B T$

Also, ideal gas system obeys ideal gas law, including the term of absolute temperature.

$PV = N k_B T$

P; pressure

V: volume

N: number of particles

k_B: Boltzmann constant

T: absolute temperature

(Definition of absolute temperature should be originated from reversible carnot engine or dS/dE)

2. Calculate the heat capacity of an ideal gas at a constant pressure $C_p = \frac{dQ}{dT}|_p$. [Hint: In this case, the gas can expand against the pressure p when absorbing the heat, which means $dQ = dE - dW$ according to the 1st law of thermodynamics. Thus, $C_p(T) = C_v(T) + \frac{d(-W)}{dT}|_p$.]

The definition of specific heat is "amount of heat consumed by the system per change in system temperature." 10/10

By thirmodynamic first rule,

$$\Delta E = \Delta Q + \Delta W = \Delta Q - P\Delta V$$

Internal energy of 3D ideal gas formula is given like this

$$E(N,T) = \frac{3}{2} k_B N T$$

Ideal gas equation is

$$PV = Nk_B T \quad , \quad P\Delta V = Nk_B \Delta T \quad (\text{at constant } P \text{ and } N)$$

So specific heat with constant pressure can be calculated like this.

$$C_P = \left(\frac{\partial Q}{\partial T}\right)_{P,N} = \left(\frac{\partial E}{\partial T}\right)_{P,N} + P\left(\frac{\partial V}{\partial T}\right)_{P,N} = \frac{3}{2} N k_B \frac{dT}{dT} + N k_B \frac{dT}{dT} = \frac{5}{2} N k_B$$

3. Explain the zeroth law of thermodynamics. 10/10

The zeroth law of thermodynamics is related to measure of temperature.

It states that if two systems are in thermal equilibrium with a third system,

they also in the thermal equilibrium each other.

This principle make the thermal equilibrium as a transitive property.

4. A differential of the form

$$df = P(x, y)dx + Q(x, y)dy$$ 10/10

is exact differential if

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

and thus $\int df$ is path-independent. Find the relation for P and Q for making df is an exact differential.

[Note: If you are not familiar with exact differential, you solve this problem after your self-study.]

We can find P and Q like this $P = \frac{\partial f}{\partial x}$ $Q = \frac{\partial f}{\partial y}$

For smooth function $f(x,y)$, the order of differentiation should be no matter.

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

The equation yields the relation between P and Q $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

Statistical Mechanics 2nd HW

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10/10

1. Let us find the entropy of an ideal gas (consisting of N particles). From the 1st law of thermodynamics, we have

$$dE = TdS - pdV.$$

Using the ideal gas law and $E = \frac{3}{2}Nk_BT$, show that

$$S(T,V) - S(T_0,V_0) = \frac{3}{2}Nk_B \ln\frac{T}{T_0} + Nk_B \ln\frac{V}{V_0} = Nk_B \ln\left\{\left(\frac{T}{T_0}\right)^{3/2}\frac{V}{V_0}\right\}.$$

Remark: (1) Note that the thermodynamic entropy is given in the form of log function! For an isothermal process, $T = T_0$, the entropy is proportional to $k_B \ln V^N$. Does this make you ring a bell? (2) Obviously, the entropy S is a state function. Its value is solely determined by the arguments of other state variables like T and V.

$E = \frac{3}{2}Nk_BT$

$$dS = \frac{1}{T}dE + \frac{P}{T}dV$$

$E = \frac{3}{2}Nk_BT$ → $dE = \frac{3}{2}Nk_B dT$

$PV = Nk_BT$ → $P = Nk_BT\frac{1}{V}$

이상기체 상태방정식

$$dS = \frac{3}{2}Nk_B\frac{1}{T}dT + Nk_B\frac{1}{V}dV$$

$$dS = \frac{3}{2}Nk_B d\ln T + Nk_B d\ln V$$

$$S(T,V) - S(T_0,V_0) = \frac{3}{2}Nk_B(\ln T - \ln T_0) + Nk_B(\ln V - \ln V_0)$$

$$= \frac{3}{2}Nk_B \ln\frac{T}{T_0} + Nk_B \ln\frac{V}{V_0}$$

$$= Nk_B \ln\left\{\left(\frac{T}{T_0}\right)^{\frac{3}{2}}\frac{V}{V_0}\right\}$$

2. Let us find the adiabatic equation for an ideal gas. Consider the adiabatic expansion/compression of an ideal gas. Now there is no heat exchange (i.e., adiabatic), the 1st law says

10/10

$$dE = dW.$$

The left-hand side is $dE = (3/2)Nk_B dT$ while the right-hand side is $dW = -pdV$. (1) By manipulating these relations, show the following adiabatic relations:

$$\left(\frac{T}{T_0}\right)^{3/2} = \left(\frac{V_0}{V}\right),$$

$$\left(\frac{T}{T_0}\right)^{5/2} = \left(\frac{p}{p_0}\right),$$

$$pV^{5/3} = \text{const.}$$

Remark: This relation shows that when the volume of an ideal gas is changed adiabatically the entropy is conserved! See the entropy of an ideal gas in Prob. 1. Does this make sense to you? Because there is no heat exchange during the adiabatic process (or transformation), the entropy is conserved.

$$\frac{3}{2}Nk_B\,dT = -P\,dV \qquad P = \frac{1}{V}Nk_BT$$

$$= -Nk_BT\frac{1}{V}dV$$

$$\frac{3}{2}\frac{1}{T}dT = -\frac{1}{V}dV$$

$$\frac{3}{2}d(\ln T) = -d(\ln V)$$

$$\frac{3}{2}(\ln T - \ln T_0) = -(\ln V - \ln V_0)$$

$$\left(\frac{T}{T_0}\right)^{\frac{3}{2}} = \frac{V_0}{V}$$

$$PV = Nk_BT$$

$$\Downarrow$$

$$PdV + VdP = Nk_B\,dT$$

$$\frac{3}{2}Nk_B\,dT = -P\,dV$$

$$= VdP - Nk_B\,dT$$

$$\frac{5}{2}Nk_B\,dT = V\,dP \qquad V = \frac{1}{P}Nk_BT$$

$$= Nk_B\frac{1}{P}dP$$

$$\frac{5}{2}d(\ln T) = d(\ln P)$$

$$\left(\frac{T}{T_0}\right)^{\frac{5}{2}} = \left(\frac{P}{P_0}\right)$$

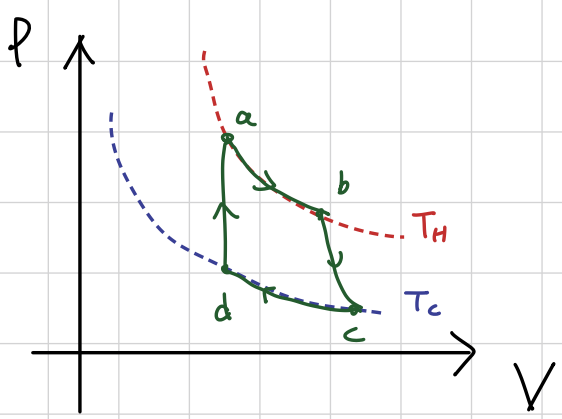
$$\left(\frac{P}{P_0}\right)\left(\frac{V}{V_0}\right)^{\frac{5}{3}} = \left(\frac{T}{T_0}\right)^{\frac{5}{2}}\left(\frac{T_0}{T}\right)^{\frac{5}{2}} = 1$$

$$PV^{\frac{5}{3}} = \text{constant}$$

3. Explicitly show that for a Carnot engine, the integral of one Carnot cyclic process gives $\oint dQ/T = 0.$

[The adiabatic expansion/compression of an ideal gas you studied in Prob. 2 is used in this exercise. Once you solve Prob. 2, use this result directly when solving this problem.]

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$a \to b$ 등온 팽창

$dQ = -dW = PdV$

$dQ = TdS$

$\int_a^b \frac{1}{T} dQ = \frac{1}{T_H} \int_a^b P dV = \frac{T_H}{T_H} N k_B \int_a^b \frac{1}{V} dV = N k_B \ln \frac{V_b}{V_a}$

$b \to c$ 단열 팽창

단열, $dQ = 0$. 적분식도 0

앞선 문제의 관계식을 쓰면, $\left(\frac{T_c}{T_H}\right)^{\frac{3}{2}} = \frac{V_b}{V_c}$

$c \to d$ 등온 압축

$\int_c^d \frac{1}{T} dQ = N k_B \int_c^d \frac{1}{V} dV = N k_B \ln \frac{V_d}{V_c}$

$d \to a$ 단열 압축

$dQ = 0$

$\left(\frac{T_H}{T_c}\right)^{\frac{3}{2}} = \frac{V_d}{V_a}$

$\oint \frac{1}{T} dQ = \int_a^b \frac{1}{T} dQ + 0 + \int_c^d \frac{1}{T} dQ + 0$

$= N k_B \ln \frac{V_b}{V_a} + N k_B \ln \frac{V_d}{V_c}$

$= N k_B \ln \frac{V_b \cdot V_c}{V_a \cdot V_d}$

$= N k_B \ln \left(\frac{T_H}{T_c}\right)^{\frac{3}{2}} \cdot \left(\frac{T_c}{T_H}\right)^{\frac{3}{2}}$

$= N k_B \ln(1)$

$= 0.$

4. Find the Kelvin and Clausius' statements of the 2nd law.

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Kelvin statement

Single heat reservoir 에서 열을 흡수하여
그것을 전부 일로 전환하며, 그 이외 효과는 전혀 없는
cyclic process 는 존재하지 않는다.

Clausius statement

저온에서 고온 물체로 열을 옮기는 것 외에
다른 효과 없는 cyclic process 는 있을 수 없다.

5. (1) In the lecture, we defined various thermodynamic potentials using the Legendre transformation. Provide a mathematical explanation on the Legendre transformation $g(p) = f(x) - xp$ with $p = \frac{\partial f}{\partial x} = f'(x)$. Show that the function g is given by

$$g(p) = f(f'^{-1}(p)) - f'^{-1}(p)p.$$

This means that the existence of the inverse function of $f'(x)$ and the one-to-one mapping to p is crucial.

(2) Let $f(x) = x^2$ and $p = f'(x)$. Find the function $g = f(x) - xp$ as a function of p. Also show that the inverse transformation $f = g + xp$ successfully finds the original function $f(x)$.

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(1)

$g(p) = f(x) - px$ 이므로, 문제의 식과 비교하면,

$f(x) = f(f'^{-1}(p))$, $x = f'^{-1}(p)$ 임을 보이면 된다.

$x = f'^{-1}(p)$ 를 확인해 보자. 이는 $p = f'(x)$ 의 다른 형태 일 것이다.

... 그저 역 함수의 성질, $F(a) = b \iff b = F^{-1}(a)$ 를 이용 하면 된다

F 가 f', a 가 x, b 가 p 이다.

$p = f'(x) \iff x = f'^{-1}(p)$

$f(x) = f(f'^{-1}(p))$ 임은 $x = f'^{-1}(p)$ 에서 자연스럽게 따라 나온다.

(2) $f(x) = x^2$ $p = f'(x)$

$f'(x) = 2x$, $p = 2x$, $x = \frac{p}{2}$, $f(x) = \frac{1}{4}p^2$, $xp = 2x^2 = \frac{1}{2}p^2$

$g(p) = f(x) - xp = \frac{1}{4}p^2 - \frac{1}{2}p^2 = -\frac{1}{4}p^2$

$f(x) = g(p) + xp = -x^2 + 2x^2 = x^2$

Statistical Mechanics 3rd HW

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1. Let us find the statement of the 2nd law in terms of the enthalpy H. Consider an isobaric system (등압계) where p is const and $dp = 0$. (Of course, the volume of the isobaric system can fluctuate under the constant pressure p.)

(1) For this system, show that

$$dH = d(E + pV) = dQ + \sum_i dW_i$$

10/10

where dQ is the amount of heat exchanged with an environment, and $dW_i (= f_i dX_i)$ is a reversible work except for the component of pdV. For instance, $dW = \mu dN$ if the system allows the change of the particle number ($dN \neq 0$). If there is no reversible work performed (i.e., $\sum_i dW_i = 0$), *the enthalpy for an isobaric system is the same as the heat exchanged, i.e.,* $dH = dQ$.

$dE = TdS - PdV + \sum_i f_i dx_i$

$dE = dQ - PdV + \sum_i dW_i$

$dH = dE + d(PV)$

$= dQ - PdV + \sum_i dW_i + PdV + VdP$

$= dQ + VdP + \sum_i dW_i$

because we assum a isobaric system, $dP = 0$

$dH = dQ + \sum_i dW_i$

if $\sum_i dW_i = 0$, then $dH = dQ$

(2) Now we further assume the situation that the isobaric system is adiabatic. In this case, $dQ = 0$ and then the enthalpy is given by

$$dH = \sum_i dW_i.$$

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Considering the 2nd law, one can generalize this equality relation to

$$dH \leq \sum_i \delta W_i.$$

$\sum_i dW_i \leq \sum_i \delta W_i$

In this expression, δW refers to a work applied irreversibly or reversibly (and we know the sign $dW_{rev} < \delta W_{irr}$). Make your own argument for this inequality relation.

Therefore, for the isobaric, adiabatic system, when there is no work exchange with the environment, we have

$$dH \leq 0.$$

For such systems, the enthalpy decreases and finally reaches the minimum value (the equilibrium condition). At this state, $dH = 0$ and $H = H_{\min}$.

Clausis inequality 를 사용하면, $\delta Q \leq TdS$ (등호는 $\delta Q = dQ_{rev}$)

$dH = \delta Q + VdP + \sum_i dW_i \leq TdS + VdP + \sum_i dW_i$

$dp = 0$ 을 쓰면,

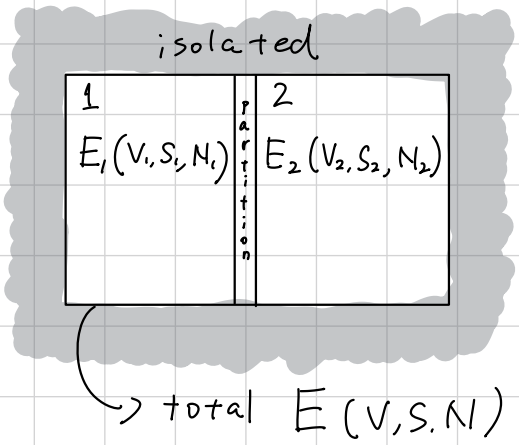
$dH = \delta Q + \sum_i dW_i \leq TdS + \sum_i dW_i$

Adiabatic 은 reversible adiabatic, $dS = 0$ 으로 해석하면,

$dH \leq \sum_i dW_i$

2. (lecture) Using the 2nd law, find the criteria for thermal, mechanical, and chemical equilibrium. Your task is to relax the constraints imposed on the wall partitioning the isolated system into two sub-systems, where the wall allows exchanges of heat, volume, and particles.

10/10



$$V_1 + V_2 = V$$
$$E_1 + E_2 = E$$
$$S_1 + S_2 = S$$
$$N_1 + N_2 = N$$
$$dV_1 = -dV_2$$
$$\vdots$$

양측 system의 entropy는, $dS_i = \frac{1}{T_i} dE_i + \frac{P_i}{T_i} dV_i - \frac{\mu_i}{T_i} dN_i$

전체 system의 entropy 변화는

$$dS = dS_1 + dS_2 = \left(\frac{1}{T_1} - \frac{1}{T_2}\right) dE_1 + \left(\frac{P_1}{T_1} - \frac{P_2}{T_2}\right) dV_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2}\right) dN_1$$

이제 partition에 constraints를 없애보자.

Adiabatic 제약을 없애서 thermal equilibrium을 정의해보자.

$dV_1 = 0$, $dN_1 = 0$, 이고 E_1과 E_2는 변할 수 있는 상태.

엔트로피 최대, $dS = 0$ 을 위해서는 $\left(\frac{1}{T_1} - \frac{1}{T_2}\right) dE = 0$ 이어야 한다.

$T_1 = T_2$ 에 도달하면 엔트로피가 최대.

접촉한 두 system의 온도가 같아지는 것을 thermal equilibrium 이라 한다.

이제 partition이 움직일 수 있게 하여 mechanical equilibrium의 조건을 찾아보자.

$\left(\frac{P_1}{T_1} - \frac{P_2}{T_2}\right) dV_1 = 0$ 이면 엔트로피 최대.

$\frac{P_1}{T_1} - \frac{P_2}{T_2} = 0$ 이어야 한다. $T_1 = T_2$ 이면, $P_1 = P_2$ 이어야 한다.

온도가 같은, 서로 접촉한 두 system의 압력이 같아지는 걸 mechanical equilibrium 이라 한다.

이제 partition이 입자를 통과할 수 있게 한다.

$\left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2}\right) dN_1 = 0$ 이어야 하므로, $\frac{\mu_1}{T_1} = \frac{\mu_2}{T_2}$ 여야 한다.

$T_1 = T_2$ 이라면, $\mu_1 = \mu_2$, 둘의 chemical potential이 같아지는게 chemical equilibrium 이다.

3. Find the chemical potential of an ideal gas based on the following steps. (1) According to the Gibbs-Duhem relation, the chemical potential is given by

$$d\mu(p,T) = -\frac{S}{N}dT + \frac{V}{N}dp.$$

Using the entropy of an ideal gas that we obtained in our previous homework, the above relation is identified to

$$d\mu(p,T) = -k_B\left(S_0 + \ln\left\{\left(\frac{T}{T_0}\right)^{5/2}\frac{P_0}{P}\right\}\right)dT + k_BT\frac{dP}{P}.$$

(2) Initially, the ideal gas was prepared such that its chemical potential is $\mu_0 = \mu(p_0, T_0)$. Now our aim is to calculate the chemical potential $\mu(p,T)$ of an ideal gas via the following quasi-static transformation: (a) $(p_0,T_0) \to (p_0,T)$ (isobaric), (b) $(p_0,T) \to (p,T)$ (isothermal). Performing the integral under this condition, show that

$$\mu(p,T) = \mu_0 - k_BT\ln\left\{\left(\frac{T}{T_0}\right)^{5/2}\frac{P_0}{P}\right\} + \left(\frac{5}{2} - S_0\right)k_B(T - T_0).$$

(3) Recall that the internal energy is a first-order homogeneous function

$$E = TS - pV + \mu N.$$

Note that we know all the thermodynamic quantities of the ideal gas. Plugging them all in the above equation, show that

$$\mu_0 = \left(\frac{5}{2} - S_0\right)k_BT_0.$$

(4) Using the result in (3), finally show that the chemical potential of an ideal gas at (p,T) is

$$\mu(p,T) = k_BT\left[\frac{\mu_0}{k_BT_0} - \ln\left\{\left(\frac{T}{T_0}\right)^{5/2}\frac{P_0}{P}\right\}\right]$$

where $\mu_0 = \mu(p_0,T_0)$ is the chemical potential at the reference state. (Note that the chemical potential difference is important).

지난 과제 →

$$\frac{S}{N} = k_B\left[S_0 + \ln\left(\left(\frac{T}{T_0}\right)^{\frac{5}{2}}\left(\frac{P}{P_0}\right)\right)\right]$$

10/10

Ideal gas equation 에 의해,

$$\frac{V}{N} = \frac{k_BT}{P}$$

두 식을 $d\mu = -\frac{S}{N}dT + \frac{V}{N}dP$ 에 대입

$$d\mu = -k_B\left[S_0 + \ln\left(\left(\frac{T}{T_0}\right)^{\frac{5}{2}}\left(\frac{P}{P_0}\right)\right)\right]dT + \frac{k_BT}{P}dP$$

(2) path integral

10/10

(a) Isobaric $(P_0, T_0) \rightarrow (P_0, T)$

압력이 같기에, $\frac{P_0}{P} = 1$, $dP = 0$

$$\Delta\mu_a = -k_B \int_{T_0}^{T} \left[S_0 + \frac{5}{2} \ln \frac{T'}{T_0} \right] dT'$$

$\int \ln\left(\frac{T'}{T_0}\right) dT' = T' \ln\left(\frac{T'}{T_0}\right) - T'$ 을 이용.

$$\Delta\mu_a = -k_B S_0 (T - T_0) - \frac{5}{2} k_B \left[T \ln \frac{T}{T_0} - T + T_0 \right]$$

(b) Isothermal $(P_0, T) \rightarrow (P, T)$

$dT = 0$ 이므로, 간단해 진다.

$$\Delta\mu_b = k_B T \int_{P_0}^{P} \frac{dP'}{P_0} = k_B T \ln \frac{P}{P_0} = \frac{5}{2} k_B T \ln \left(\frac{P}{P_0}\right)^{\frac{2}{5}}$$

$$\mu(P,T) = \mu_0 + \Delta\mu_a + \Delta\mu_b = \mu_0 - k_B T \ln \left[\left(\frac{T}{T_0}\right)^{\frac{5}{2}} \frac{P_0}{P} \right] + \left(\frac{5}{2} - S_0 \right) k_B (T - T_0)$$

(3) $E = TS - pV + \mu N$ 에 $E = \frac{3}{2} N k_B T$, $pV = N k_B T$ 대입,

10/10

$$\frac{3}{2} N k_B T = TS - N k_B T + \mu N$$

$$\mu = \frac{1}{N} \left(\frac{5}{2} N k_B T - TS \right) = \frac{5}{2} k_B T - T \frac{S}{N}$$

$\frac{S}{N} = k_B \left[S_0 + \ln \left(\left(\frac{T}{T_0}\right)^{\frac{5}{2}} \left(\frac{P}{P_0}\right) \right) \right]$ 에 $S = S_0$, $T = T_0$ 대입,

$\frac{S_0}{N} = k_B S_0 + k_B \ln(1) = k_B S_0$ 따라서,

$\mu_0 = \frac{5}{2} k_B T_0 - T_0 k_B S_0 = \left(\frac{5}{2} - S_0 \right) k_B T_0$ 이다.

(4) (3)의 결과를 (2)에 대입

10/10

$$\mu(P,T) = \mu_0 + \frac{\mu_0}{T_0} (T - T_0) - k_B T \ln \left[\left(\frac{T}{T_0}\right)^{\frac{5}{2}} \frac{P_0}{P} \right]$$

$$= \mu_0 \frac{T}{T_0} - k_B T \ln \left[\left(\frac{T}{T_0}\right)^{\frac{5}{2}} \frac{P_0}{P} \right]$$

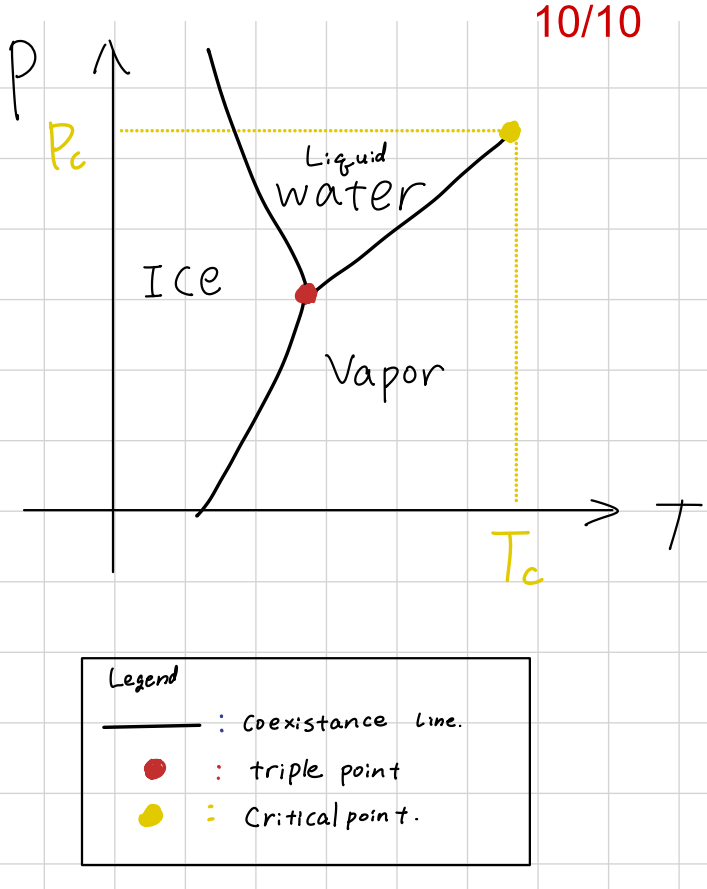
$$= k_B T \left[\frac{\mu_0}{k_B T_0} - \ln \left(\left(\frac{T}{T_0}\right)^{\frac{5}{2}} \frac{P_0}{P} \right) \right]$$

Statistical Mechanics 4th HW

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1. (lecture) Draw the phase diagram of H_2O molecules in the P-T plane. Explain the coexistence line, critical point, and triple point.

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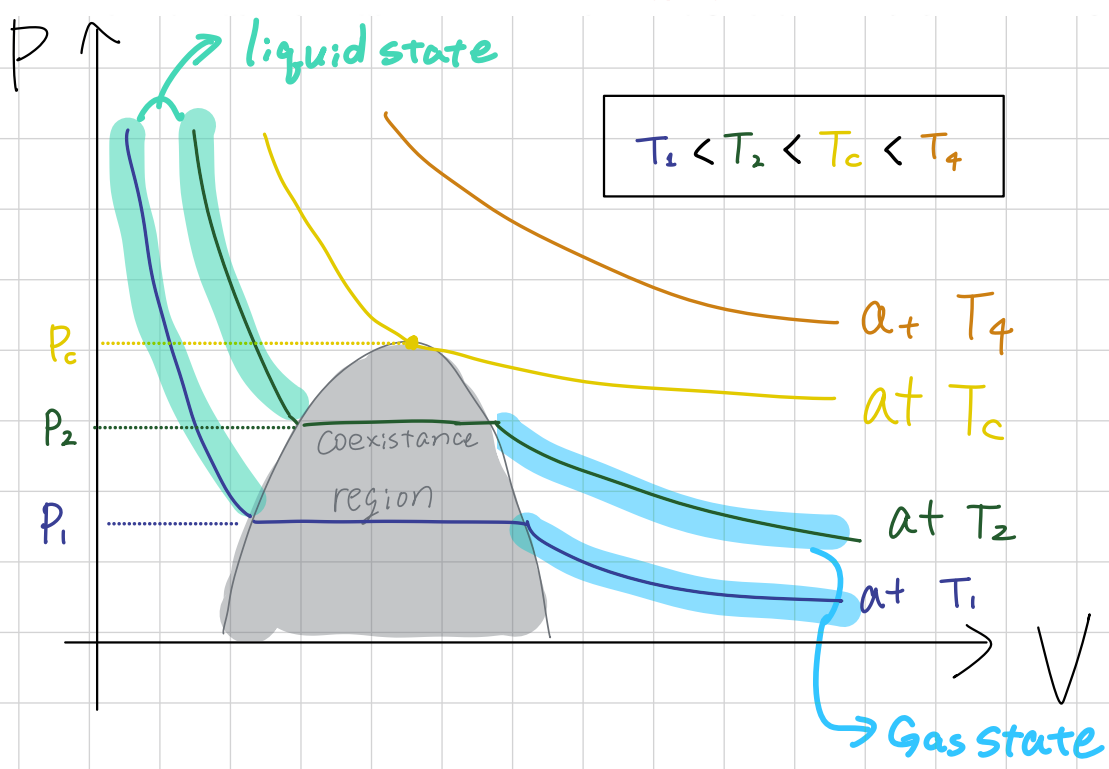
Coexistance line : 두 phase가 동시에 공존하는 압력과 온도이다.
Coexistance line 위의 (P,T) 라면, 상황에 따라 시스템이 다른 부피와 엔트로피를 가질 수 있다.

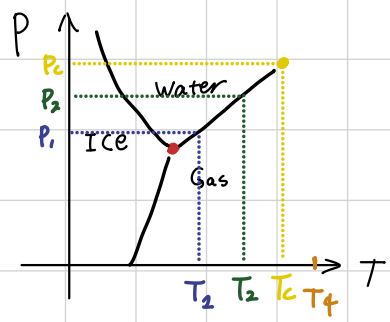
Critical point: Coexistance line이 끊기는 지점이다.
이 이상의 압력, 혹은 온도에서는 liquid와 gas의 구분이 사라진다.

Triple point : 세 가지의 phase가 공존할 수 있는 압력과 온도이다.
Gibbs phase rule에 따라,
PT diagram에서 오로지 한 점으로 존재한다.

2. (lecture) Repeat to draw the P-V isotherms of the water-vapor fluid at several temperatures $T \leq T_c$ where T_c is the critical temperature.

10/10





회색으로 칠한 영역같이, Vapor와 water가 공존하는 영역에서는 압력이 일정하게 유지된다.

T_c 에서는 $\left(\frac{\partial P}{\partial V}\right)_{T_c}=0$, $\left(\frac{\partial^2 P}{\partial V^2}\right)_{T_c}=0$ 이 되기 때문에 변곡점의 기울기가 0.

line이 coexistance region에 접하여,
다른 V에서 P가 같은 현상이 없어지는 온도이다.

T_c 이상의 온도에서는 supercritical fluid가 된다.

3. Let us solve the Clausius-Clapeyron equation and find the coexisting curve of the water-vapor. The Clapeyron equation is given by

$$\frac{dp}{dT} = \frac{\Delta Q_{\text{lat}}}{T\Delta v}.$$

Q 와 v 를 입자당 열과 부피로 해석.

Let us assume that the vapor behaves as an ideal gas and $\Delta v = v_{\text{gas}} - v_{\text{wat}} \approx v_{\text{gas}}$. Show that the above equation is written as

$$\frac{dp}{dT} = \frac{p}{k_B T^2}\Delta Q_{\text{lat}}.$$

> 물의 부피가 vapor 에 비해 무시 가능하다 가정.

Now assume that the latent heat per particle does not depend on pressure. Show that the coexisting curve is

$$p(T) = p_0(T_0)\exp\left\{-\frac{\Delta Q_{\text{lat}}}{k_B}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right\}$$

$Q_{lat} \not\propto p$

10/10

where p_0 is the saturated vapor pressure at temperature T_0.

Ideal gas equation: $PV = Nk_BT$

$$V_{(P,T)} = Nk_B\frac{T}{P}$$

$$\Delta v = v_{gas} = k_BT/P$$

$$\frac{dp}{dT} = \frac{\Delta Q_{lat}}{T\Delta v} = \frac{P}{k_BT^2}\Delta Q_{lat}$$

ΔQ_{lat} 가 P와 T 에 대해 상수라고 가정.

$$\frac{1}{P}dP = \frac{1}{k_BT^2}\Delta Q_{lat}\,dT$$

$$\left[\ln P\right]_{T_0}^{T} = \frac{-\Delta Q_{lat}}{k_B}\left[\frac{1}{T}\right]_{T_0}^{T}$$

$$\ln\frac{P(T)}{P(T_0)} = -\frac{\Delta Q_{lat}}{k_B}\left(\frac{1}{T} - \frac{1}{T_0}\right)$$

$$P(T) = P_0(T_0)\exp\left\{-\frac{\Delta Q_{lat}}{k_B}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right\}$$

4. From the Van der Waals equation of state, find the critical temperature T_c, pressure p_c, and volume V_c. Using the rescaled temperature, pressure, and volume, write down the Van der Waals equation of corresponding states.

10/10

$$\left(P+\frac{a}{v^2}\right)(v-b)\simeq kT,$$

교과서 식.

$$P(v)=\frac{kT}{v-b}-\frac{a}{v^2}$$

Critical point $\longrightarrow$ $P(V)$ 의 변곡점의 기울기가 0 이 되는 순간.

$$\max\left(\frac{\partial P}{\partial V}\right)=0$$ 이것을 만족하는 v_c 를 찾아라.

equation 을 V로 미분.

$$P(v)=\frac{kT}{v-b}-\frac{a}{v^2}$$

$$\frac{dP}{dv}=-kT(v-b)^{-2}+2av^{-3}$$

$\max\left(\frac{dP}{dv}\right)$ 를 찾기 위해 한번 더 미분. $\frac{d^2P}{dv^2}=2kT(v-b)^{-3}-6av^{-4}$

$\left(\frac{d^2P}{dv^2}\right)\Big|_{v_m}=0$ 을 만족하는 v_m 은, $v_m^4(v_m-b)^{-3}=\frac{3a}{kT}$

$\left(\frac{dP}{dv}\right)\Big|_{v_m}=-kT(v_m-b)^{-2}+2av_m^{-3}=0$ 을 만족하면, $v_m=v_c$

$$\begin{cases} v_c^3(v_c-b)^{-2}=\frac{2a}{kT_c} & \cdots (1) \\ v_c^4(v_c-b)^{-3}=\frac{3a}{kT_c} & \cdots (2) \end{cases}$$

$v_c(v_c-b)^{-1}=\frac{3}{2}$ $\quad v_c=3b$ $\quad V_c=3Nb$

(1) 에 $v_c=3b$ 대입 $3^3\cdot 2^{-2}b=\frac{2a}{kT_c}$ $\quad T_c=\frac{8}{27k}\cdot\frac{a}{b}$

$$P_c=\frac{kT_c}{v_c-b}-\frac{a}{v_c^2}=\frac{8}{27}\cdot\frac{a}{b}\cdot\frac{1}{2b}-\frac{a}{9b^2}=\frac{1}{27}\cdot\frac{a}{b^2}$$ $\quad P_c=\frac{1}{27}\cdot\frac{a}{b^2}$

Rescaling $\longrightarrow$ $v=v_c\tilde{v}$, $T=T_c\tilde{T}$, $P=P_c\tilde{P}$

$\left(P+\frac{a}{v^2}\right)(v-b)=kT$ $\longrightarrow$ $\left(\frac{1}{27}\frac{a}{b^2}\tilde{P}+\frac{1}{9}\frac{a}{b^2}\cdot\frac{1}{\tilde{v}^2}\right)(3b\tilde{v}-b)=\frac{8}{27}\cdot\frac{a}{b}\tilde{T}$

$$\left(\tilde{P}+3\frac{1}{\tilde{v}^2}\right)(3\tilde{v}-1)=8\tilde{T}$$

↳ a, b 상관 없이 모든 반데르발스 기체가 같은 식을 따른다.

5. (textbook, prob 1.9) Making use of the fact that the entropy $S(N,V,E)$ of a thermodynamic system is an extensive quantity, show that

10/10

$$N\left(\frac{\partial S}{\partial N}\right)_{V,E} + V\left(\frac{\partial S}{\partial V}\right)_{N,E} + E\left(\frac{\partial S}{\partial E}\right)_{N,V} = S$$

Note that this result implies that $(-N\mu + PV + E)/T = S$, that is $N\mu = E + PV - TS$.

N, V, E, S 가 모두 extensive variable 임을 이용하자.

system size 가 ℓ 에 비례한다면,

$$\ell S(N,V,E) = S(\ell N, \ell V, \ell E)$$

양변을 ℓ 로 미분

$$S(N,V,E) = \frac{\partial \ell N}{\partial \ell}\left(\frac{\partial S}{\partial \ell N}\right)_{V,E} + \frac{\partial \ell V}{\partial \ell}\left(\frac{\partial S}{\partial \ell V}\right)_{N,E} + \frac{\partial \ell E}{\partial \ell}\left(\frac{\partial S}{\partial \ell E}\right)_{N,V}$$

$\ell = 1$ 을 대입

$$S(N,V,E) = N\left(\frac{\partial S}{\partial N}\right)_{V,E} + V\left(\frac{\partial S}{\partial V}\right)_{N,E} + E\left(\frac{\partial S}{\partial E}\right)_{N,V}$$

$$dE = \mu dN - P dV + T dS$$

$$dS = \frac{1}{T}dE - \frac{\mu}{T}dN + \frac{P}{T}dV$$

$$\left(\frac{\partial S}{\partial E}\right)_{N,V} \quad \left(\frac{\partial S}{\partial N}\right)_{E,V} \quad \left(\frac{\partial S}{\partial V}\right)_{E,N}$$

$$S = -\frac{\mu N}{T} + \frac{PV}{T} + \frac{E}{T}$$

$$TS = -\mu N + PV + E$$

Statistical Mechanics 5th HW

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1-(1) TR 에 의해, $TR : (q, p, t) \to (q, -p, -t)$ 10/10

$\{\rho, \mathcal{H}\} \to \{\mathcal{H}, \rho\} = -\{\rho, \mathcal{H}\}$ 임을 보여라.

풀이 → $\{\rho, \mathcal{H}\} = \sum_i \frac{\partial \rho}{\partial q_i}\frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial \mathcal{H}}{\partial q_i}\frac{\partial \rho}{\partial p_i}$ 일단은 해밀토니안이 TR에 불변이라 가정하자.

After time reversal,

$TR(\{\rho, \mathcal{H}\}) = \sum_i \frac{\partial \rho}{\partial q_i}\frac{\partial \mathcal{H}}{\partial(-p_i)} - \frac{\partial \mathcal{H}}{\partial q_i}\frac{\partial \rho}{\partial(-p_i)} = \sum_i \frac{\partial \mathcal{H}}{\partial q_i}\frac{\partial \rho}{\partial p_i} - \frac{\partial \rho}{\partial q_i}\frac{\partial \mathcal{H}}{\partial p_i} = \{\mathcal{H}, \rho\} = -\{\rho, \mathcal{H}\}$

$(q(t), p(t))$ 가 Hamilton equation 의 solution 이면,
$(q(-t), -p(-t))$ 또한 solution 임을 보여라.

$\dot{q}_i(t) = \frac{\partial \mathcal{H}}{\partial p_i}$ $-\dot{p}_i(t) = \frac{\partial \mathcal{H}}{\partial q_i}$ 이면,

$\dot{q}_i(-t) = \frac{\partial \mathcal{H}}{\partial(-p_i)}$, $\dot{p}_i(-t) = \frac{\partial \mathcal{H}}{\partial q_i}$ 또한 성립함을 보이자.

$\dot{q}_i(-t) = \frac{\partial \mathcal{H}}{\partial(-p_i)}$ 부터 확인, $\dot{q}(-t) = \frac{d}{dt}q(-t) = \frac{d(-t)}{dt}\frac{dq(-t)}{d(-t)} = -\frac{dq(t)}{dt} = -\dot{q}(t)$

$\frac{\partial \mathcal{H}}{\partial(-p_i)} = -\frac{\partial \mathcal{H}}{\partial p_i}$ ≡

따라서, $\dot{q}_i(-t) = \frac{\partial \mathcal{H}}{\partial(-p_i)} \iff \dot{q}(t) = \frac{\partial \mathcal{H}}{\partial p_i}$

$\dot{p}_i(-t) = \frac{\partial \mathcal{H}}{\partial q_i}$ 을 확인, 위와 같은 논리로, $\dot{p}_i(-t) = -\dot{p}_i(t)$

따라서, $\dot{p}_i(-t) = \frac{\partial \mathcal{H}}{\partial q_i} \iff -\dot{p}_i(t) = \frac{\partial \mathcal{H}}{\partial q_i}$

1-(2) 이 관계성을 비가역성의 측면에서 해석하라. 10/10

풀이 → 고전역학은 순전히 가역적인 운동밖에 설명 못한다. 시간을 거꾸로한 운동도 운동방정식의 해가 될 수 있으므로.
즉 모든 운동을 '미시적'으로 보면 다 가역적이다.
'비가역성'은 여러 입자의 운동을 뭉뚱그려 볼 때 나타난다.

2. $\frac{d\rho}{dt} = 0$ 이라는 Liouville's theorem 유도를 다시 해 보라.

10/10

풀이 →

ρ 에 대한 연속체 방정식.

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathcal{U})$$

$$\mathcal{U}(\{q_i\}, \{p_i\}, t) = \sum_i \hat{q}_i \dot{q}_i + \hat{p}_i \dot{p}_i = \sum_i \hat{q}_i \frac{\partial \mathcal{H}}{\partial p_i} - \hat{p}_i \frac{\partial \mathcal{H}}{\partial q_i}$$

phase space 에서 flow field.

여기 에서 해밀턴 방정식 $\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \sum_i \left(\dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right)$ 을 유도 할 수 있다.

∇ 에서 ∇ 연산자 정의. : $\nabla = \sum_i \left(\hat{q}_i \frac{\partial}{\partial q_i} + \hat{p}_i \frac{\partial}{\partial p_i} \right)$

$\nabla \cdot (\rho \mathcal{U}) = (\nabla \rho) \cdot \mathcal{U} + \rho (\nabla \cdot \mathcal{U})$ 벡터 항등식 이용.

$$\nabla \cdot \mathcal{U} = \sum_i \left(\frac{\partial^2 \mathcal{H}}{\partial q_i \partial p_i} - \frac{\partial^2 \mathcal{H}}{\partial p_i \partial q_i} \right) = 0$$

$$\nabla \rho = \sum_i \left(\hat{q}_i \frac{\partial \rho}{\partial q_i} + \hat{p}_i \frac{\partial \rho}{\partial p_i} \right)$$

$$\nabla \cdot (\rho \mathcal{U}) = \sum_i \left(\frac{\partial \rho}{\partial q_i} \frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial \rho}{\partial p_i} \frac{\partial \mathcal{H}}{\partial q_i} \right)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathcal{U}) = 0 \Rightarrow \frac{\partial \rho}{\partial t} + \sum_i \left(\frac{\partial \rho}{\partial q_i} \frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial \mathcal{H}}{\partial q_i} \frac{\partial \rho}{\partial p_i} \right) = \frac{d\rho}{dt} = 0$$

해밀턴 방정식을 떠올리면, 연속체 방정식은 $\frac{d\rho}{dt} = 0$ 을 의미한다는 걸 알 수 있다.

$\frac{d\rho}{dt} = 0$ 는 ρ의 총량은 시간이 지나도 변하지 않음을 의미 한다.

3-(1) 「 에서 dw 의 불변성을 (x, y, z, P_x, P_y, P_z) 에서 $(r, \theta, \varphi, P_r, P_\theta, P_\varphi)$ 으로 좌표계 변환하는 상황에서 보여라. 10/10

풀이 → in cartesian coordinate, $dw = dx\,dy\,dz\,dP_x\,dP_y\,dP_z$.

Cartesian coordinate

$x = r\sin\theta\cos\varphi \longrightarrow dx = \underset{A}{dr\sin\theta\cos\varphi} + \underset{B}{d\theta\, r\cos\theta\cos\varphi} - \underset{C}{d\varphi\, r\sin\theta\sin\varphi}$

$y = r\sin\theta\sin\varphi \longrightarrow dy = \underset{D}{dr\sin\theta\sin\varphi} + \underset{E}{d\theta\, r\cos\theta\sin\varphi} + \underset{F}{d\varphi\, r\sin\theta\cos\varphi}$

$z = r\cos\theta \longrightarrow dz = \underset{G}{dr\cos\theta} - \underset{H}{d\theta\, r\sin\theta}$

$$dx\wedge dy = (dr\sin\theta\cos\varphi + d\theta\, r\cos\theta\cos\varphi - d\varphi\, r\sin\theta\sin\varphi)\wedge(dr\sin\theta\sin\varphi + d\theta\, r\cos\theta\sin\varphi + d\varphi\, r\sin\theta\cos\varphi)$$

$$= \underbrace{(\sin\theta\cos\varphi\, r\cos\theta\sin\varphi - r\cos\theta\cos\varphi\,\sin\theta\sin\varphi)}_{=0} dr\wedge d\theta$$

$$+ (r\cos\theta\cos\varphi\, r\sin\theta\cos\varphi + r\sin\theta\sin\varphi\, r\cos\theta\sin\varphi)\, d\theta\wedge d\varphi$$

$$- (\sin\theta\sin\varphi\, r\sin\theta\sin\varphi + r\sin\theta\cos\varphi\,\sin\theta\cos\varphi)\, d\varphi\wedge dr$$

$$= (r^2\sin\theta\cos\theta)\, d\theta\wedge d\varphi - (r\sin^2\theta)\, d\varphi\wedge dr$$

$$dx\wedge dy\wedge dz = \{(r^2\sin\theta\cos\theta)\, d\theta\wedge d\varphi - (r\sin^2\theta)\, d\varphi\wedge dr\}\wedge(dr\cos\theta - d\theta\, r\sin\theta)$$

$$= \{\cos\theta\,(r^2\sin\theta\cos\theta) + r\sin\theta\,(r\sin^2\theta)\}(dr\wedge d\theta\wedge d\varphi)$$

$$= r^2\sin\theta\,(dr\wedge d\theta\wedge d\varphi)$$

$\therefore\ dx\,dy\,dz = r^2\sin\theta\, dr\,d\theta\,d\varphi$

$$\dot{x} = \dot{r}\sin\theta\cos\varphi + \dot{\theta}\, r\cos\theta\cos\varphi - \dot{\varphi}\, r\sin\theta\sin\varphi$$

$$\dot{y} = \dot{r}\sin\theta\sin\varphi + \dot{\theta}\, r\cos\theta\sin\varphi + \dot{\varphi}\, r\sin\theta\cos\varphi$$

$$\dot{z} = \dot{r}\cos\theta - \dot{\theta}\, r\sin\theta.$$

$$T = \frac{m}{2}|\mathbb{V}|^2 = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right)$$ kinetic energy

$$L = T$$

일반화 운동량을 정의한다.

$$P_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} =$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2\sin^2\theta\,\dot{\phi}$$

$$\mathbb{V} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\hat{\phi}$$

$$\mathbb{P} = m\mathbb{V}$$

$$P_r = \mathbb{P}\cdot\hat{r}$$

$$P_\theta = r\,\mathbb{P}\cdot\hat{\theta}$$

$$P_\phi = r\sin\theta\,\mathbb{P}\cdot\hat{\phi}$$

따라서, $\mathbb{P}$ 는

$$\mathbb{P} = P_r\hat{r} + \frac{1}{r}P_\theta\hat{\theta} + \frac{1}{r\sin\theta}\hat{\phi}$$

$P_x\ P_y\ P_z$ 와 $P_r\ P_\theta\ P_\phi$ 의 변환을 생각하면

$R(\theta,\phi)$ 가 rotation operator 일 때,

$$\begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix} = R(\theta,\phi)\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/r & 0 \\ 0 & 0 & 1/r\sin\theta \end{pmatrix}\begin{pmatrix} P_r \\ P_\theta \\ P_\phi \end{pmatrix}$$

Jacobideterminant를 구하자면,

$$\det J = \det\frac{\partial(P_x,P_y,P_z)}{\partial(P_r,P_\theta,P_\phi)} = \underbrace{\det(R)}_{\text{회전이니까 1이다.}}\det\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/r & 0 \\ 0 & 0 & 1/r\sin\theta \end{pmatrix}$$

$\therefore \det J = \frac{1}{r^2 \sin\theta}$

$\therefore dP_x dP_y dP_z = \frac{1}{r^2 \sin\theta} dP_r dP_\theta dP_\varphi$

$$dx dy dz\, dP_x dP_y dP_z = \left(r^2 \sin\theta\, dr d\theta d\varphi\right)\left(\frac{1}{r^2 \sin\theta} dP_r dP_\theta dP_\varphi\right)$$

$$= dr d\theta d\varphi\, dP_r dP_\theta dP_\varphi$$

3-(2) (2) The foregoing result seems to contradict the intuitive notion of "equal weights for equal solid angles," because the factor $\sin\theta$ is invisible in the expression for $d\omega$. Show that if we average out any physical quantity, whose dependence on p_θ and p_φ comes only through the kinetic energy of the particle, then as a result of integration over these variables we do indeed recover the factor $\sin\theta$ to appear with the subelement $(d\theta d\varphi)$.

10/10

$$T = \frac{P_r^2}{2m} + \frac{P_\theta^2}{2mr^2} + \frac{P_\phi^2}{2mr^2 \sin^2\theta}$$

Observable A의 평균은, 상태 분포 ρ에 대해,

$$\langle A \rangle = \int A\rho\, d\omega = \int A\rho\, dr d\theta d\varphi\, dP_r dP_\theta dP_\varphi$$

$A\rho$가 어떤 함수, $f(r, \theta, \varphi, P_r, T)$ 꼴이라고 치자.

$\langle A \rangle$ 에서 $dP_\theta\, dP_\phi$ 적분만 계산하면,

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} A\rho\, dP_\theta dP_\varphi = \iint f\left(r, \theta, \varphi, P_r, \frac{1}{2m}\left(P_r^2 + \frac{P_\theta^2}{r^2} + \frac{P_\varphi^2}{r^2 \sin\theta}\right)\right) dP_\theta dP_\varphi$$

치환적분, $P_\theta = ru$, $P_\phi = r\sin\theta\, v$. $\quad (P_\theta, P_\phi) \to (u, v)$

$$\frac{dP_\theta}{du} = r \;,\; \frac{dP_\phi}{dv} = r\sin\theta \;, \quad dP_\theta dP_\phi = r^2 \sin\theta\, du dv.$$

$$T = \frac{1}{2m}\left(P_r^2 + u^2 + v^2\right)$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\left(r, \theta, \varphi, P_r, \frac{1}{2m}\left(P_r^2 + \frac{P_\theta^2}{r^2} + \frac{P_\varphi^2}{r^2 \sin\theta}\right)\right) dP_\theta dP_\varphi = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f\left(r, \theta, \varphi, P_r, \frac{1}{2m}\left(P_r^2 + u^2 + v^2\right)\right) r^2 \sin\theta\, du dv.$$

$\therefore \langle A \rangle \propto r^2 \sin\theta \iint f\left(r, \theta, \varphi, P_r, \frac{1}{2m}(P_r^2 + u^2 + v^2)\right) du dv$

Statistical Mechanics 6th HW

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1. Ideal gas: Consider an ideal gas system of N particles with the Hamiltonian $\mathcal{H} = \sum_{i=1}^{3N} \frac{p_i^2}{2m}$. Complete the calculation omitted in the lecture.

(1) Show that the number of microstates is ω/h^{3N} where $\omega = \frac{(2\pi mE)^{3N/2}}{(3N/2)!}\left(\frac{V}{h^3}\right)^N$.

(2) The entropy of the ideal gas is

$$S = k_B \ln \Omega = Nk_B \left\{ \frac{3}{2} + \ln \left[\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right] \right\}$$

or

$$S = Nk_B \left\{ \frac{3}{2} + \ln \left[\frac{V}{h^3} \left(\frac{4\pi mE}{3N} \right)^{3/2} \right] \right\}.$$

(3) The internal energy: $\frac{\partial S}{\partial E}|_{V,N}$.

(4) The equation of state: $\frac{\partial S}{\partial V}|_{E,N}$.

중복으로 들어갔어.

↓

나중에 오류 수정 공지 올려주셨다.

늦은 시간까지 수고하십니다. 감사합니다!

(1) Ideal gas의 microstate 개수 세기

위치와 운동량은 독립적이므로.

ω = (position space 에서 state 개수) · (momentum space에서 state 개수)

$= \omega_q \cdot \omega_p$

position space 에서 state 개수 구하기 : 다른 particle 위치 사이 correlation 없음.

$\omega_q = ($ 한 입자가 차지할 수 있는 위치들 $)^N = V^N$

momentum space에서 state 개수 구하기 ⇒ momentum space 에서 같은 내부 에너지 E를 가지는 면의 면적

momentum space : 3N dimension

i : index of molecule

$E = \sum_i \frac{|p_i|^2}{2m}$

$\left(\sum_i |p_i|^2 \right)^{\frac{1}{2}} = \sqrt{2mE}$ → momentum space 에서 한 벡터의 노름.

즉, 구하고자 하는 건 " 3N 차원 공간에서 반지름이 $\sqrt{2mE}$ 인 구의 표면적 ~ 구의 부피 "

N 차원 공간에서 구의 부피 공식

$V_d(r) = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)} r^d$

r: 반지름, d: dimension.

$\omega_p = V_{3N}(\sqrt{2mE}) = \frac{\pi^{\frac{3N}{2}}}{(3N/2+1)!}(2mE)^{\frac{3N}{2}}$

$= \frac{2^{\frac{3N}{2}} \cdot \pi^{\frac{3N}{2}}}{(3N/2+1)!}(mE)^{\frac{3N}{2}}$

N이 아주 크기에, $N \simeq N \pm 1$ 이라 근사.

$\simeq \frac{(2\pi mE)^{\frac{3N}{2}}}{(3N/2)!}$

$\therefore$ # of microstats $= \Omega = \frac{\omega_q \cdot \omega_p}{h^3} = \frac{(2\pi mE)^{\frac{3N}{2}}}{(3N/2)!}\left(\frac{V}{h^3}\right)^N$

원래 N개의 구분 안 되는 입자라면 N! 의 multiplicity 를 나누어야 함

(2) The entropy of ideal gas.

$$\frac{S}{k_B} = \ln\Omega = \ln \frac{(2\pi mE)^{\frac{3N}{2}}}{(3N/2)!}\left(\frac{V}{h^3}\right)^N$$

$$= \frac{3N}{2}\ln(2\pi mE) + N(\ln V - 3\ln h) - \ln((3N/2)!)$$

$$= \frac{3N}{2}\left(\ln\frac{3N}{2} - 1\right)$$

Stirling approximation

$\ln N! = N(\ln N - 1)$ for Large N

$$= \frac{3N}{2}\left[\ln(2\pi mE) + \frac{2}{3}\ln V - 2\ln h - \ln\frac{3N}{2} + 1\right]$$

$$= \frac{3N}{2}\left[\ln\frac{4\pi mE}{3N}\cdot\frac{V^{\frac{2}{3}}}{h^2} + 1\right]$$

$$\therefore S = k_B N\left[\frac{3}{2} + \ln\left\{\frac{V}{h^3}\cdot\left(\frac{4\pi mE}{3N}\right)^{\frac{3}{2}}\right\}\right]$$

(3) The internal energy (temperature) of ideal gas.

$$dE = TdS - PdV + \mu dN$$

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}$$

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{\partial}{\partial E}\left(k_B N\frac{3}{2}\ln E\right)\Big|_{V,N}$$

$$= \frac{3Nk_B}{2E} = \frac{1}{T}$$

$$E = N\cdot\frac{3}{2}k_B T.$$

$\ln(2\pi mE) = \ln E + \ln 2\pi m$

미분 계산 시 필요 없음.

(4) Ideal gas state equation.

$$dE = TdS - PdV \Rightarrow dS = \frac{1}{T}(dE + PdV)$$

$$\left(\frac{\partial S}{\partial V}\right)_{N,E} = \frac{P}{T}$$

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{\partial}{\partial V}(k_B N \ln V) = \frac{1}{V} N k_B = \frac{P}{T} \Rightarrow \boxed{PV = N k_B T}$$

(lecture) In this problem set, you derive the Maxwell-Boltzmann (MB) distribution for the equilibrium velocity distribution of gas molecules from the Boltzmann equation. At equilibrium (for $t \to \infty$), $\frac{dH}{dt} = 0$, where $H(t)$ is the Boltzmann H function. The condition for equilibrium can be obtained from the Boltzmann equation as

$$f_1(\mathbf{x}, \mathbf{p}_1, t) f_1(\mathbf{x}, \mathbf{p}_2, t) = f_1(\mathbf{x}, \mathbf{p}_1', t) f_1(\mathbf{x}, \mathbf{p}_2', t),$$

where f_1 is the 1-particle density function. At equilibrium, the 1-particle density reaches its equilibrium distribution, i.e., $f_1(\mathbf{x}, \mathbf{p}, t \to \infty) = f_1^0(\mathbf{p})$. Note that f_1^0 has no dependence on $\mathbf{x}$. Thus, the above condition can be rewritten as:

$$\ln f_1^0(\mathbf{p}_1) + \ln f_1^0(\mathbf{p}_2) = \ln f_1^0(\mathbf{p}_1') + \ln f_1^0(\mathbf{p}_2').$$

From this relation, it can be inferred that f_1^0 is a function of conserved quantities upon collision, such as momentum, energy, and particle number. We assume that f_1^0 is given in the form of

$$f_1^0(\mathbf{p}) = C \exp\left[-A\mathbf{p}^2\right].$$

Now let us determine the constants A and C with some known parameters and find the MB distribution by performing the calculation below.

헷갈릴 뻔,

2. For the particle density $n(= N/V)$, the 1-particle density has the relation

$$\int d^3\mathbf{p} f_1^0(\mathbf{p}) = n.$$

Normalization으로 C를 찾는 과정.

Show that f_1^0 is given by

$$f_1^0(\mathbf{p}) = n\left(\frac{A}{\pi}\right)^{3/2} \exp\left[-A\mathbf{p}^2\right].$$

Gausian integral

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{a}} dx = \sqrt{a\pi}$$

x, y, z가 동등.

$$\int d^3\mathbf{p}\, f_1^0(\mathbf{p}) = C\int d^3\mathbf{p} \exp(-A\mathbf{p}^2) = C\left[\int dp_x C \exp(-A p_x^2)\right]^3 = C\cdot\left(\frac{\pi}{A}\right)^{\frac{3}{2}} = n \qquad C = n\left(\frac{A}{\pi}\right)^{\frac{3}{2}}$$

3. Let the average energy of a gas molecule be ϵ, which is obtained from f_1^0 as

$$\epsilon = \frac{\int f_1^0(\mathbf{p})\,(\mathbf{p}^2/2m)\,d^3\mathbf{p}}{\int d^3\mathbf{p} f_1^0(\mathbf{p})}.$$

에너지로 A를 결정

n

By performing this integral, show that f_1^0 is given by

$$f_1^0(\mathbf{p}) = n\left(\frac{3}{4\pi\epsilon m}\right)^{3/2} \exp\left[-\frac{3}{4m\epsilon}\mathbf{p}^2\right].$$

$$\int d^3p\, f_1^0(p)\,\frac{p}{2m} = C\int d^3p\, \exp\left[-A(p_x^2+p_y^2+p_z^2)\right]\frac{p_x^2+p_y^2+p_z^2}{2m}$$

$$= \frac{3C}{2m}\int d^3p\, \exp\left[-A(p_x^2+p_y^2+p_z^2)\right] p_x^2$$

$$= \frac{3C}{2m}\left(\int dp_y \exp\left(-Ap_y^2\right)\right)\left(\int dp_z \exp\left(-Ap_z^2\right)\right)\left(\int dp_x \exp(-Ap_x^2)p_x^2\right)$$

$\int dp_y \exp(-Ap_y^2) = \sqrt{\frac{\pi}{A}}$, $\int dp_z \exp(-Ap_z^2) = \sqrt{\frac{\pi}{A}}$

Gaussian integral

$$\int_{-\infty}^{\infty} x^{2n} e^{-x^2/a^2}\,dx = \sqrt{\pi}\,\frac{a^{2n+1}(2n-1)!!}{2^n}$$

$$= \frac{3C}{2m}\cdot\frac{\pi}{A}\int dp_x \exp(-Ap_x^2)p_x^2$$

$\int dp_x \exp(-Ap_x^2)p_x^2 = \pi^{\frac{1}{2}}\cdot\frac{1}{2}\cdot A^{-\frac{3}{2}}$

$$= \frac{3C}{4m}\cdot\pi^{\frac{3}{2}}A^{-\frac{5}{2}}$$

$$= \frac{3}{4m}\pi^{\frac{3}{2}}A^{-\frac{5}{2}}\cdot n\left(\frac{A}{\pi}\right)^{\frac{3}{2}}$$

$$= \frac{3n}{4m}A^{-1} = n\varepsilon$$

$$A = \frac{3}{4m\varepsilon}$$

4. The pressure exerted by the gas molecules is P. The pressure, P, is the average force exerted by the gas molecules per unit area, which is

$$P = \int_0^\infty dp_x \int_{-\infty}^\infty dp_y \int_{-\infty}^\infty dp_z \, (2p_x) \, \frac{p_x}{m} f_1^0(\mathbf{p}).$$

By performing the integral, show that

$$P = \frac{2}{3} n\epsilon = \frac{2}{3}\frac{N}{V}\epsilon.$$

$-\infty$ 여기는 대신 0

$$P = \int_0^\infty dp_x \int_{-\infty}^\infty dp_y \int_{-\infty}^\infty dp_z \; C \exp(-A\mathbf{p}^2) \frac{2}{m} p_x^2$$

$$= \frac{2C}{m} \left(\int_0^\infty dp_x \exp(-A p_x^2) p_x^2 \right) \left(\int_{-\infty}^\infty dp_y \exp(-A p_y^2) \right) \left(\int_{-\infty}^\infty dp_z \exp(-A p_z^2) \right)$$

$$\frac{1}{2}\int_{-\infty}^\infty dp_x \exp(-A p_x^2) p_x^2 = \frac{1}{4}\pi^{\frac{1}{2}} A^{-\frac{3}{2}}$$

$$\pi A^{-1}$$

$$= \frac{1}{2m} C \cdot \pi^{\frac{3}{2}} A^{-\frac{5}{2}} \qquad C = n \cdot A^{\frac{3}{2}} \pi^{-\frac{3}{2}}$$

$$= \frac{n}{2mA} \qquad A = \frac{3}{4m\varepsilon}$$

$$P = \frac{2}{3} n\varepsilon$$

5. From Prob. 4, it is found that

$$PV = \frac{2}{3}N\epsilon = NkT.$$

Thus, we find the famous relation

$$\epsilon = \frac{3}{2}kT.$$

Plugging this relation into the f_1^0 in Prob. 3, show the f_1^0 is given by the MB distribution

$$f_1^0(\mathbf{p}) = \frac{n}{(2\pi mkT)^{3/2}} \exp\left[-\frac{\mathbf{p}^2}{2mkT}\right].$$

$$f_1^0(p) = n\left(\frac{3}{4\pi\varepsilon m}\right)^{\frac{3}{2}} \exp\left(-\frac{3}{4m\varepsilon}p^2\right)$$

$$\Downarrow \quad \varepsilon = \frac{3}{2}kT$$

$$f_1^0(p) = \frac{n}{(2\pi mkT)^{3/2}} \exp\left(-\frac{3}{4m\varepsilon}p^2\right)$$

정이 그래로 고전적인 경로로 이동한다.

이게 리우빌 과 같을때

뇌정량을 풀이 중.

볼정상의 정의

influx : $J_+(x, p_1, t) = ?$. outflux $J_-(x, p_1, t) = ?$

Detailed balance 이용